

EVENT PREP

AMD AI DevDay India 2026

Prerequisites & Prep Guide

Everything to learn before Aug 8 — AMD's story, the AI stack, and every workshop with hands-on projects

"You don't need to be an AI expert to walk in. You need the map: what AMD is betting on, what each session actually means, and a small project behind each one — so the demos click instead of blur."

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-

Part A — AMD: The Company Behind the Event

Before you learn the tech, understand the host. Everything on stage is AMD proving one thing: that you can build serious AI without NVIDIA. Knowing the story makes every demo land harder.

A1. Who AMD Is (in one breath)

Simple definition: AMD (Advanced Micro Devices) is a ~57-year-old American chip company that designs the "brains" inside computers — the CPU (general thinking), the GPU (massively parallel math), and now the NPU (an efficient AI co-processor). It designs the chips; factories like TSMC in Taiwan physically build them (AMD is "fabless").

Telugu: AMD ante computer lopala unde brain chips (CPU, GPU, NPU) design chese company. Vaalla daggara sonta factory ledu — design matram chestaru, TSMC laantivi chips ni physically tayararu chestayi. Idi "fabless" model. Ee event motham AMD cheppadam: "NVIDIA lekunda kuda meeru AI build cheyyochu" ani.

Analogy: AMD is the architect who draws the building; TSMC is the construction crew that pours the concrete. AMD's whole pitch at this event: their "buildings" (chips) can now run the heaviest AI workloads that used to need NVIDIA.

Why it matters for you: the workshops all run on AMD silicon — Ryzen AI laptops for the local-model sessions, Instinct/Radeon GPUs behind the training and video demos. The vocabulary in Part C is what the speakers will assume you know.

A2. The History — 1969 to Now

AMD's arc is a genuine underdog story — near-bankruptcy twice, then a stunning comeback. You don't need every date, but the shape explains *why they're hungry for AI*.

Era	What happened	Why it matters
1969	Founded by Jerry Sanders + 7 others, months after Intel	Born as the scrappy #2 to Intel
1975–90s	Made Intel-compatible x86 chips (second-source)	Learned to survive in Intel's shadow
1999–2006	Athlon / Athlon 64 beat Intel; bought ATI (graphics) for \$5.4B	Gained the GPU business it now uses for AI
2006–2014	Bulldozer flops, near-bankruptcy, sold its fabs (→ GlobalFoundries)	Became "fabless"; nearly died
2014	Dr. Lisa Su becomes CEO	The turnaround begins
2017	Zen / Ryzen launches — competitive CPUs again	AMD is back; stock and share soar
2020	Ryzen + EPYC take huge market share from Intel	Cash to invest in AI
2022	Acquires Xilinx (\$50B, FPGA/adaptive) + Pensando (networking/DPU)	Buys the "adaptive compute" + networking pieces
2023–2025	Buys Nod.ai (AI compiler), Silo AI (\$665M), ZT Systems (~\$4.9B)	Assembles a full AI software + systems stack
2023→	Instinct MI300 launches; ROCm matures	AMD's serious answer to NVIDIA's data-center AI GPUs

Telugu: AMD rendu sarlu almost bankrupt ayyindi. 2014 lo Lisa Su CEO ga vachaka, 2017 lo "Ryzen" CPUs tho comeback kottaru. Aa comeback tho vachina dabbu + confidence tho ippudu AI meeda pedda bet vestunnaru. Anduke ee event lo antha energy — vaallu prove cheyyali NVIDIA ni challenge cheyyagalamu ani.

The one name to know: Dr. Lisa Su — CEO since 2014, an electrical engineer (MIT PhD), widely credited with the turnaround. If the keynote references "our journey," this is it.

A3. The Product Families

AMD makes four kinds of "compute engines." Every session touches at least one.

Family	Type	What it's for	You'll see it in...
Ryzen	Consumer CPU (+ NPU)	Laptops & desktops; Ryzen AI adds an NPU for on-device AI	Vibe Coding (local model)
EPYC	Server CPU	Data-center CPUs that feed the GPUs	Behind every training demo
Threadripper	Workstation CPU	High-core creator/dev workstations	Content-creation context
Radeon	Consumer GPU	Gaming + increasingly local AI (fine-tune, image/video)	Unsloth, ComfyUI
Instinct	Data-center GPU	The heavy AI accelerators (training + large inference)	RL/LLM training talk
Ryzen AI (XDNA NPU)	NPU	Low-power AI co-processor inside Ryzen laptops	Local agent demos
Versal / Xilinx	FPGA / Adaptive	Reconfigurable chips (telecom, auto, edge)	"Sovereign / edge AI" context

Rule of thumb: **CPU** = a few very smart workers doing tasks in order. **GPU** = thousands of simple workers doing the same math at once (perfect for AI). **NPU** = a small, power-sipping GPU-like unit tuned only for AI, so your laptop battery survives.

A4. Where AMD Chips Actually Run

Not abstract — you've used AMD without knowing:

- **PlayStation 5 & Xbox Series X/S** — custom AMD CPU+GPU (semi-custom business).
- **Frontier & El Capitan supercomputers** — the world's fastest, on AMD EPYC + Instinct.
- **Microsoft Azure, Meta, Oracle Cloud** — deploy AMD Instinct for AI. Meta alone reportedly runs 173,000+ MI300X units; 7 of the 10 largest AI companies now use Instinct.
- **Most gaming laptops & many data centers** — Ryzen / EPYC.

Telugu: Meeru already AMD vaadaru — PS5, Xbox lo AMD chip ne. World's fastest supercomputers kuda AMD meede. Ippudu Meta, Microsoft laantivi AI kosam AMD Instinct GPUs vaadutunnaru. So "AMD AI lo kotha" kaadu — vaallu already big players.

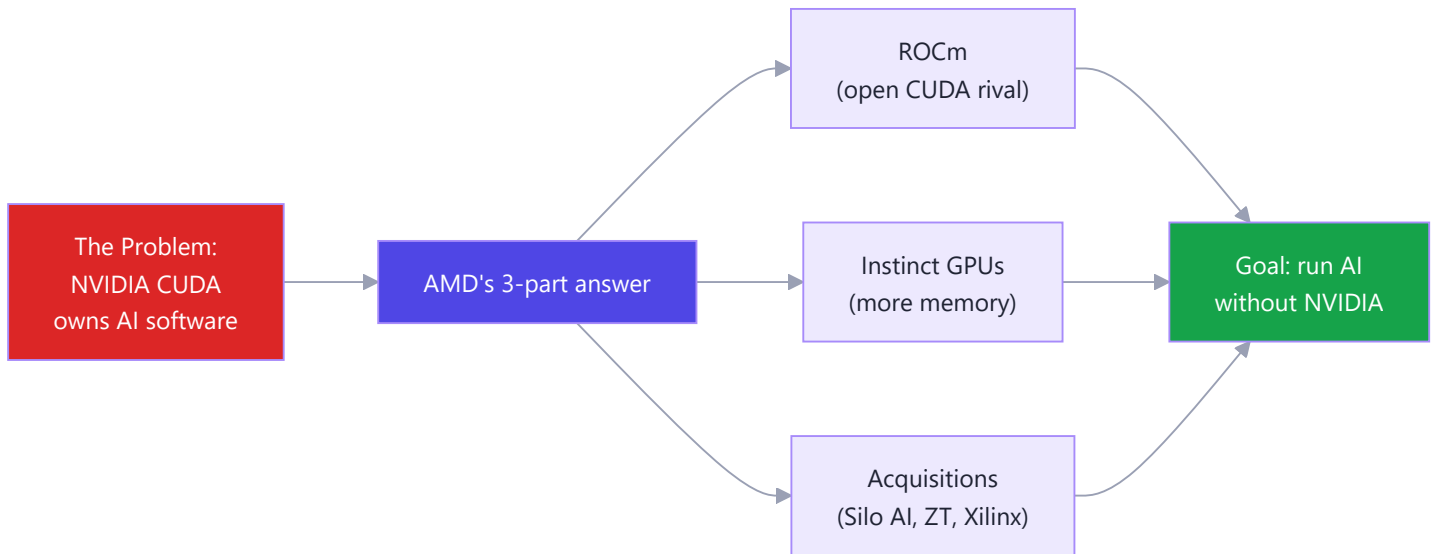
A5. How & Why AMD Entered AI

This is the heart of the event. The story has three beats:

Beat 1 — The CUDA moat. For ~15 years, AI ran almost exclusively on NVIDIA GPUs, because NVIDIA built **CUDA** — the software layer every AI framework (PyTorch, TensorFlow) was written against. Hardware was never the whole game; the *software ecosystem* was NVIDIA's real moat. Buy an AMD GPU in 2018 and most AI code simply wouldn't run.

Beat 2 — AMD's answer: ROCm + Instinct. AMD built **ROCm** (their open-source CUDA equivalent) and the **Instinct** GPU line with one killer advantage: **memory**. AI models are huge; the more high-speed memory (VRAM/HBM) on one GPU, the bigger the model you can run without splitting it. The MI300X shipped with far more memory than NVIDIA's flagship, and MI350X pushes to **288GB HBM3e** — a real edge for running giant models.

Beat 3 — Buying the missing pieces. Hardware alone wasn't enough, so AMD went shopping: **Silo AI** (Europe's largest private AI lab — model-building talent), **Nod.ai** (AI compilers), **ZT Systems** (rack-scale systems know-how), **Xilinx & Pensando** (adaptive compute + networking). The goal: a *full-stack* AI offering — chips, software, systems — not just a GPU.



Telugu: NVIDIA balam hardware kaadu — "CUDA" ane software. Prapancham lo AI code antha CUDA meeda raasaru, so AMD GPU meeda run avvadu. AMD daaniki javabu: (1) ROCm — CUDA laantidi kaani open-source, (2) Instinct GPUs — ekkuva memory (pedda models fit avtayi), (3) companies ni konnaru (Silo AI, ZT Systems) — missing skills kosam. Motham lakshyam: NVIDIA lekunda AI run cheyyagalagatam.

Why the event exists at all: AMD needs *developers* to actually use ROCm and Ryzen AI. A chip with no software community is dead. DevDay is AMD seeding that community in India — hence free workshops, awards, and "From Campus to Code."

A6. AMD vs NVIDIA in 2026 — The Honest Picture

Don't walk in as a fanboy of either. The grown-up view:

Dimension	NVIDIA	AMD
Software maturity	CUDA — 18 yrs, everything supports it	ROCm 7 — much better, still catching up
Raw training speed	Usually wins (e.g. Blackwell)	Slightly behind, but close on tuned workloads
Memory per GPU	Less (e.g. B200 ~180GB)	More (MI350X ~288GB HBM3e)
Cost / power efficiency	Premium price	Often better \$/performance and perf/watt
Local / laptop AI	Weaker consumer NPU story	Strong — Ryzen AI Max, unified memory
Ecosystem lock-in	Very high (CUDA)	Open (ROCm, open-source friendly)

Telugu: NVIDIA speed lo, software lo inka mundu. Kaani AMD rendu chotla gelustundi: (1) okka GPU meeda ekkuva memory — pedda models run avtayi, (2) laptop meeda local AI — Ryzen AI Max chala strong. Price/power kuda AMD ki better ekkuva sarlu. Balanced ga chudandi — "AMD better" kaadu, "AMD ippudu real choice" ani.

Your one-liner for the day: *"NVIDIA still wins raw speed and software maturity; AMD wins on memory-per-GPU, cost/watt, and local on-device AI — and ROCm 7 closed most of the gap."* Say that to any speaker and you'll sound informed.

Part B — The Foundations You Need First

The event's own requirement: "LLM training knowledge, agentic systems experience, Python/Jupyter familiarity." This Part gets you to that bar. If you've done Phase 12 of your roadmap (Claude API + agents), you're already halfway.

B1. Python & Jupyter — The Lab Bench

Simple definition: **Python** is the language of AI — every framework (PyTorch, Hugging Face, vLLM, Unsloth) is Python-first. **Jupyter** (and its cloud twin **Google Colab**) is a notebook where you write and run code in small cells and see output inline — perfect for experiments.

Telugu: AI antha Python lo ne. Jupyter/Colab ante code ni chinna chinna "cells" ga raasi, prati cell run chesi result ven/ventane chudatam. Experiments ki super — oka cell marchi malli run cheyyochu, motham program malli start cheyyakkarledu.

You don't need to be a Python expert — you need to *read* it and run cells. Know these:

```
# Variables, lists, dicts, functions – the 20% you'll see everywhere
name = "AMD"
gpus = ["MI350X", "Radeon", "Ryzen AI"]      # list
spec = {"memory_gb": 288, "type": "HBM3e"}    # dict (like JS object)

def summarize(items):                          # function
    for i, x in enumerate(items):              # loop with index
        print(f"{i}: {x}")                    # f-string (like JS template literal)

summarize(gpus)
```

```
# Installing a package inside a notebook cell (the ! runs a shell command)
!pip install transformers torch
```

JS → Python cheat (you're a web dev): `const` → just assign; `console.log` → `print`; `${x}` → `f"{x}"`; `array.map` → list comprehension `[f(x) for x in arr]`; `{}` object → `dict`; `null` → `None`; `true/false` → `True/False`.

Do before the event: open colab.research.google.com, run 10 cells, get comfortable. That's genuinely enough.

B2. What an LLM Actually Is

Simple definition: A Large Language Model is a giant math function trained on much of the internet to do one thing astonishingly well: **predict the next token** (word-piece) given the text so far. Everything else — chat, code, reasoning — emerges from doing that at massive scale.

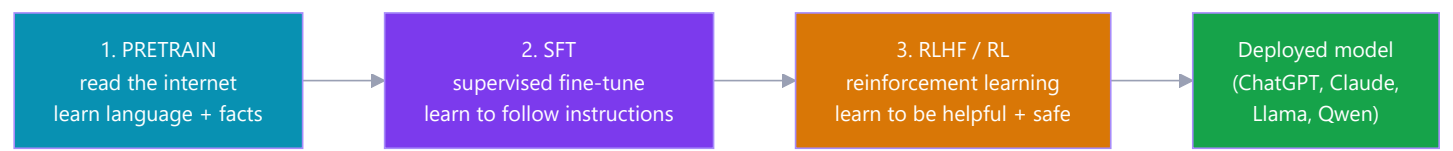
Telugu: LLM ante oka pedda math function — internet antha chadivi "next word enti?" ani guess chestundi. Chat, code, reasoning — anni aa okka pani (next token predict cheyyadam) nunche puttayi, chala pedda scale valla. Magic kaadu, patterns matrame.

Analogy: Autocomplete on your phone, but trained on trillions of words and with billions of tunable knobs (**parameters**). It doesn't "look up" answers — it computes the most likely continuation.

You already met this in your roadmap's Claude API phase. The event assumes you know: LLMs are *trained*, they have *parameters*, they run on *GPUs*, and they can be *fine-tuned* and *served*. The next sections unpack each.

B3. How LLMs Are Trained (Pretrain → SFT → RLHF)

This single pipeline is the backbone of **three** sessions (the RL talk, the Unsloth fine-tuning workshop, and the vibe-coding model). Learn it once here.



Stage	Plain English	Analogy	Data used
1. Pretraining	Read trillions of words; learn grammar, facts, reasoning patterns	A student reading every book in the library	Raw internet text
2. SFT (Supervised Fine-Tuning)	Show it good "question → answer" examples so it follows instructions	Giving the student worked examples	Curated Q&A pairs
3. RLHF / RL	Reward good answers, penalize bad ones, so it <i>prefers</i> helpful, safe replies	A coach giving thumbs-up / thumbs-down	Human preferences / reward signals

Telugu: Model ni 3 steps lo train chestaru. (1) Pretrain — internet antha chadivistaru, language nerchukuntadi. (2) SFT — manchi "prashna → javabu" examples chupistar, instructions follow cheyyadam nerchukuntadi. (3) RLHF — manchi javabu ki reward, chedu javabu ki penalty istaru, so helpful ga behave chestundi. Ee 3 steps ee event lo 3 sessions ki base — gurtu pettuko.

Key insight for the day: Pretraining costs millions and only big labs do it. **You** — and most of the workshops — work at stages 2 and 3: *fine-tuning* (Unsloth) and *RL* (the HF talk) on already-pretrained open models like Llama or Qwen. That's why open models matter so much.

B4. The Vocabulary — Tokens, Parameters, Quantization, Context

Four words the speakers will use constantly. Nail these and you'll follow everything.

Term	Simple meaning	Why you care at the event
Token	A word-piece (~¾ of a word). Models read/write in tokens	Pricing, speed ("tokens/sec"), context limits are all in tokens
Parameters	The tunable "knobs" (weights). 7B = 7 billion of them	Model size; bigger = smarter but needs more memory
Quantization	Storing those knobs in fewer bits (16-bit → 4-bit) to shrink the model	This is <i>how</i> a 30B model fits in a laptop; central to local-AI demos
Context window	How many tokens the model can "see" at once (its short-term memory)	Limits how much code/doc you can feed an agent

Telugu: Token = word mukka (~¾ word). Parameters = model lopala unde knobs (7B ante 700 crore knobs). Quantization = aa knobs ni takkuva bits lo store cheyyatam (16-bit → 4-bit), so model chinnaga ayyi laptop lo fit avtundi. Context = model okesari entha text choodagalado (short-term memory). Ee 4 words speakers pdeddi sarlu vaadataru.

The quantization "aha": A 7B model in full 16-bit precision needs ~14GB of memory. Quantized to 4-bit, it needs ~4GB — now it runs on a laptop. **This is the entire reason local AI (Ryzen AI, Lemonade) is possible.** When a speaker says "4-bit quantized Qwen," they mean "we shrank it so it runs on this laptop's memory."

B5. GPU Compute 101 — VRAM, TOPS, Why AI Needs GPUs

Simple definition: AI math is millions of the *same* small operations (matrix multiplies) that can all happen at once. A **CPU** has a few fast cores (does things in sequence); a **GPU** has thousands of small cores (does them in parallel) — a perfect match. An **NPU** is a tiny GPU-like unit tuned only for AI, sipping power.

Telugu: AI lo same chinna math (matrix multiply) million sarlu jarugutundi — anni okesari cheyyochu. CPU ki konni fast cores (sequence lo), GPU ki veyyila chinna cores (anni okesari) — AI ki perfect. NPU ante chinna GPU laantidi, AI kose, takkuva power tho.

Two numbers you'll hear:

- **VRAM / memory (GB):** how big a model + its data can fit *on the GPU*. Run out and the model won't load (or must be split across GPUs). This is AMD's headline advantage — MI350X's 288GB, Strix Halo's 128GB unified memory.
- **TOPS (Tera-Operations/sec):** raw AI throughput. The Ryzen AI Max NPU is ~50 TOPS. Higher = faster inference.

Memory is the gatekeeper. A common beginner shock: "I have a fast GPU but the model won't run." Almost always it's *not enough VRAM*, not speed. That's why "unified memory" (Part C5) is a big deal — it lets the model use system RAM as VRAM.

B6. CUDA vs ROCm vs HIP — The Platform War in One Page

The single most important thing to understand about *why this event exists*.

Layer	NVIDIA world	AMD world	What it is
The GPU	GeForce / Instinct-competitor	Radeon / Instinct	The hardware
Low-level API	CUDA	ROCm (+ HIP)	How code talks to the GPU
Frameworks	PyTorch, vLLM... (CUDA build)	PyTorch, vLLM... (ROCm build)	What you actually write

- **CUDA** = NVIDIA's proprietary software layer. The reason NVIDIA dominated: everyone wrote for CUDA.
- **ROCm** = AMD's *open-source* equivalent. ROCm 7 now officially supports PyTorch, vLLM, and llama.cpp — meaning most modern AI code runs on AMD with little or no change.
- **HIP** = a thin translation layer: write once, run on *both* AMD and NVIDIA. AMD even ships **HIPIFY** to auto-convert CUDA code to HIP.

Telugu: CUDA = NVIDIA sonṭa software. Andaru CUDA meeda raasaru, so NVIDIA gelichindi. ROCm = AMD di, open-source, same pani chestundi. HIP = translator laantidi — okesari raasi AMD, NVIDIA rendintlo run cheyyochu. ROCm 7 ippudu PyTorch, vLLM support chestundi — so AMD meeda modern AI code run avtundi. Ee point event motham ki heart.

Your takeaway: When a speaker says "it just works on ROCm now," they mean the old "AMD can't run AI code" problem is largely solved. That claim — and how true it is — is the subtext of the whole day.

Part C — The AMD AI Stack You'll See on Stage

Now the AMD-specific names. Each maps to a session or a demo laptop/rack in the room.

C1. ROCm — AMD's AI Software Platform

Simple definition: ROCm (Radeon Open Compute) is AMD's full open-source software stack that lets AI frameworks run on AMD GPUs — the direct answer to CUDA. Think "drivers + libraries + compilers" that make PyTorch talk to a Radeon or Instinct GPU.

Telugu: ROCm ante AMD GPUs meeda AI run cheyyadaniki kavalsina software motham — CUDA ki direct javabu, open-source. PyTorch ni AMD GPU tho matladinchedi idi.

- **ROCm 7** (current): AMD claims $\sim 3.5\times$ inference and $\sim 3\times$ training gains over ROCm 6; officially supports PyTorch, vLLM, llama.cpp on Instinct + select Radeon cards.
- **HIP + HIPIFY:** write portable GPU code; auto-convert CUDA $\rightarrow$ HIP.
- **Where you meet it:** invisibly under every training/serving demo. If someone shows `torch.cuda.is_available()` returning `True` on an AMD box, that's ROCm doing the work (AMD keeps the `cuda` name for drop-in compatibility — a small, telling detail).

C2. Instinct — Data-Center AI GPUs

Simple definition: Instinct is AMD's line of *data-center* GPUs built only for AI/HPC (no display output) — the chips that train large models and serve them at scale. The rivals to NVIDIA's H100/H200/B200.

Model	Memory	Notable
MI300X	192GB HBM3	Meta deployed 173k+; broke NVIDIA's monopoly
MI325X	256GB HBM3e	Memory bump
MI350X	288GB HBM3e, $\sim 8\text{TB/s}$	2026 flagship; competes with NVIDIA B200

Telugu: Instinct = AMD data-center AI GPUs, sirf AI kose (screen output undadu). Ivi pedda models train chestayi. MI350X ki 288GB memory — NVIDIA kanna ekkuva, so pedda models okka GPU meeda fit avtayi.

Where you meet it: the "Scaling RL for LLMs" (Hugging Face) talk and any training/benchmarks. When they show cluster training, it's EPYC CPUs + Instinct GPUs (+ ZT Systems racks).

C3. Radeon — Consumer GPUs for AI

Simple definition: Radeon is AMD's mainstream gaming GPU line — but with ROCm support, mid/high-end Radeon cards now double as affordable AI cards for *fine-tuning* and *image/video generation* at home.

Telugu: Radeon = gaming GPUs, kaani ROCm valla ivi ippudu inti daggara AI ki kuda pani chestayi — fine-tuning, image/video generation ki cheap option.

Where you meet it: the **Unsloth fine-tuning** and **ComfyUI video** workshops target Radeon-class GPUs — the "you can do this at home" message.

C4. Ryzen AI & the NPU (XDNA)

Simple definition: "Ryzen AI" = Ryzen laptop chips that include a dedicated **NPU** (Neural Processing Unit, AMD's **XDNA** architecture) — a low-power engine that runs AI *on the laptop*, no cloud, no big GPU, without draining the battery.

Telugu: Ryzen AI ante laptop chip lo pratyekam ga oka NPU (XDNA) untundi — chinna, power-saving AI engine. Cloud avasaram ledu, battery kuda taggadu. Meeru offline AI run cheyyochu.

- **NPU vs GPU vs CPU:** the NPU does steady, efficient AI (a local chatbot, live transcription) while sipping power; the GPU does heavy bursts; the CPU coordinates.
- **Where you meet it:** every "local model" / "on-device" demo — especially Vibe Coding.

C5. Ryzen AI Max+ 395 "Strix Halo" — Star of the Local Demos

Simple definition: The Ryzen AI Max+ 395 (codename **Strix Halo**) is AMD's most powerful laptop/mini-PC chip for local AI — it fuses a strong CPU, a big integrated GPU, and an NPU, all sharing one huge pool of **unified memory** so it can run models that normally need a data-center GPU.

The specs that matter (why the room will "ooh"):

Spec	Value	Why it matters
CPU	16 Zen 5 cores	Fast general compute
GPU	Radeon 8060S, 40 CU	Real GPU muscle, no discrete card
NPU	XDNA 2, ~50 TOPS	Efficient always-on AI
Unified memory	up to 128GB LPDDR5X, ~256 GB/s	Up to 96GB usable as VRAM
Model reach	Quantized models up to ~200B params	Runs models a normal laptop can't dream of

Telugu: Strix Halo (Ryzen AI Max+ 395) = AMD laptop chip lo strongest for local AI. CPU + GPU + NPU anni kalisi oke pedda memory pool (128GB varaku) share chestayi. Id "unified memory" — andulo 96GB varaku VRAM la vaadochu. So normal laptop lo run avvani pedda models (30B, even bigger) ee laptop lo local ga run avtayi. Vibe Coding session lo ide star.

The "unified memory" trick — the key idea: On a normal PC, the GPU has its *own* small memory (say 8–16GB), separate from system RAM. A big model won't fit. Strix Halo makes CPU, GPU, and NPU *share one 128GB pool* — so the GPU can borrow tons of memory and load a 30B+ model that a \$1,500 discrete GPU can't. **That's the whole reason it can run big models locally** — and the core of the Vibe Coding demo.

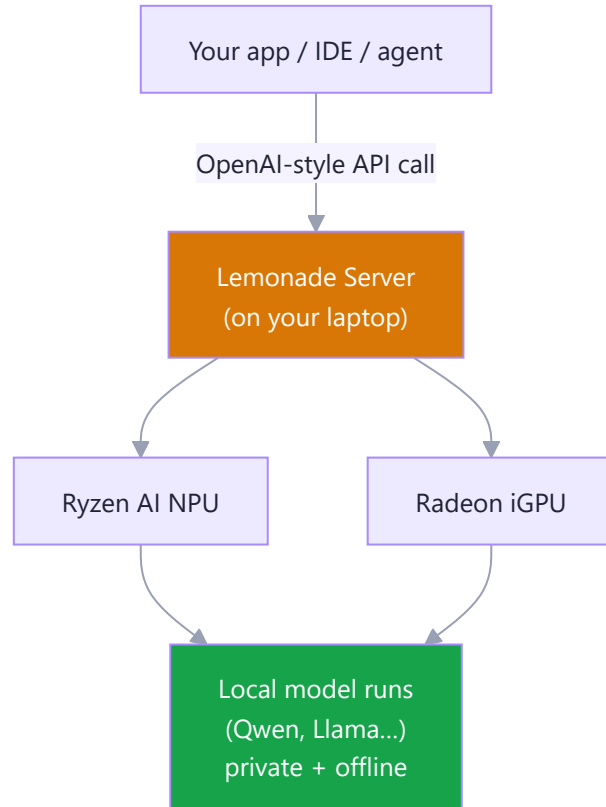
C6. Lemonade Server & GAIA — Local Models on Ryzen AI

Simple definition: **Lemonade Server** is AMD's open-source local LLM server — it runs models on the Ryzen AI NPU + iGPU and exposes them through a standard **OpenAI-compatible API**, so your apps talk to a *local* model

exactly like they'd talk to ChatGPT. **GAIA** is AMD's open-source framework for building local *agents* on top of Lemonade.

Telugu: Lemonade Server ante mee Ryzen AI laptop lo model ni run chesi, daanini "OpenAI laanti API" ga icche software. So mee code local model tho matladutundi — cloud tho matladinatte. GAIA daani meeda agents build cheyyadaniki AMD framework. Vibe Coding session lo iv` vaadataru.

- **OpenAI-compatible = plug-and-play:** point any tool that speaks the OpenAI API at <http://localhost:...> and it uses your *local* model — private, offline, free per-token.
- **Where you meet it:** Vibe Coding (Lemonade + Qwen Coder) and any local-agent demo (GAIA).



Part D — Concept: Scaling RL for LLMs (Hugging Face)

Session: "Scaling RL for LLMs: Infrastructure, Environments, and Training" — Hugging Face. This is the most research-heavy talk. Here's the mental model so it lands.

D1. What Reinforcement Learning Adds to an LLM

Simple definition: Reinforcement Learning (RL) trains a model by *trial, reward, and adjustment* rather than by copying examples. For LLMs, it's stage 3 (Part B3): after the model can follow instructions, RL teaches it to *prefer* answers humans (or a checker) rate as better.

Telugu: RL ante examples copy cheyyatam kaadu — model pryatnimchi, reward/penalty tho nerchukuntundi. LLM lo idi 3rd step: instructions follow cheyyaka, "manchi javabu" ni prefer cheyyadam nerpistundi. Ela? Manchi javabu ki reward, chedu daaniki penalty.

Analogy: SFT is a student memorizing worked solutions. RL is the student then *taking practice tests* and adjusting from the score. That "learn from the score" step is what makes ChatGPT feel helpful instead of just knowledgeable.

D2. RLHF, Reward Models, PPO → GRPO

The alphabet soup, decoded:

Term	Plain English
RLHF	RL from <i>Human Feedback</i> — humans rank answers; the model learns to produce top-ranked ones
Reward model	A second model trained to <i>predict</i> those human scores automatically (so you don't need a human for every answer)
PPO	Proximal Policy Optimization — the classic (complex, memory-heavy) RL algorithm behind original ChatGPT
GRPO	Group Relative Policy Optimization — a newer, <i>simpler, cheaper</i> method (popularized by DeepSeek) that skips a lot of PPO's baggage
RLVR	RL from <i>Verifiable Rewards</i> — for math/code, the "reward" is just: did the answer pass the test? No human needed

Telugu: RLHF = humans javabulani rank chestaru, model top-ranked vaatini nerchukuntundi. Reward model = aa human scores ni predict chese 2nd model (prati sari human avasaram ledu). PPO = classic RL algorithm (complex). GRPO = kotta, simple, cheap version (DeepSeek famous chesindi). Math/code ki RLVR — reward simple: test pass ayindaa leda ane.

The 2026 story the talk will ride: RL used to be PPO-only, fiddly, and expensive. **GRPO** and **verifiable rewards** made RL dramatically cheaper and reproducible — sparking the "reasoning model" wave (models that think step-by-step and self-check). Hugging Face's **TRL** library is the standard open toolkit for all of this.

D3. Why "Scaling RL" Is Genuinely Hard

The talk's subtitle — "Infrastructure, Environments, and Training" — flags the three hard parts:

- **Infrastructure:** RL needs the model to *generate* answers (inference) AND *learn* (training) in a tight loop across many GPUs — orchestrating that at scale is an engineering beast. (This is where AMD Instinct + ROCm + ZT Systems racks come in.)
- **Environments:** you need tasks with checkable outcomes (math problems, coding tests, games). Building good, diverse, cheat-proof "environments" is half the battle.
- **Training stability:** RL can collapse (the model finds a way to game the reward — "reward hacking"). Keeping it stable is an art.

Telugu: "Scaling RL" enduku kashtam? (1) Infrastructure — model generate cheyyali + learn cheyyali, rendu oke loop lo, chala GPUs meeda. (2) Environments — check cheyyagalige tasks kavali (math, code tests). (3) Stability — model reward ni cheat cheyyochu ("reward hacking"), daanini aapadam kashtam. Ee 3 challenges e talk motham.

D4. Real-World Scenarios

- **ChatGPT / Claude helpfulness** — RLHF is why they refuse bad requests and format nicely.
- **DeepSeek-R1 reasoning** — GRPO + verifiable rewards produced a top open reasoning model cheaply.
- **Code models** — RLVR: reward = "unit tests pass," directly improving real coding ability.
- **Agents** — RL is increasingly used to train agents that plan multi-step tasks (ties into Part F).

D5. Mini-Project: See RLHF with TRL

Goal: run one tiny RL fine-tune so the concepts stop being abstract. ~1 hour on Google Colab (free). You'll watch a model learn from a reward.

Step 1 — Open a free GPU notebook. [Google Colab](#) → Runtime → Change runtime type → GPU.

Step 2 — Install the toolkit.

```
!pip install trl transformers datasets peft
```

Step 3 — Load a tiny model + a reward idea. Use a small model (e.g. a 135M–1B model so it fits free tier).

Define a trivial, *verifiable* reward — e.g. "reward longer, more detailed answers" or "reward answers containing a keyword." This mirrors RLVR: reward is a Python function, no humans.

```
# Pseudocode shape of a GRPO run in TRL – read for the *shape*, not to memorize
from trl import GRPOTrainer, GRPOConfig
from datasets import load_dataset

def reward_fn(completions, **kw):
    # verifiable reward: +1 per completion that includes the word "because"
    return [1.0 if "because" in c else 0.0 for c in completions]

trainer = GRPOTrainer(
    model="Qwen/Qwen2.5-0.5B-Instruct",
    reward_funcs=reward_fn,
    args=GRPOConfig(num_generations=4, max_steps=50),
    train_dataset=load_dataset("trl-lib/tldr", split="train[:200]"),
)
trainer.train()
```

Step 4 — Watch the reward rise. Over steps, the average reward should climb — the model *learns* to include "because" (i.e., to explain). That's RL in miniature: no examples of the right answer, just a score it optimizes toward.

Step 5 — Reflect (this is the point). You just did what the HF talk scales to thousands of GPUs. Now their "infrastructure / environments / stability" challenges will mean something concrete.

What to say at the talk: *"When you scale GRPO, is the bottleneck the generation step or the gradient step — and how does Instinct's memory help?"* Instant credibility.

Part E — Concept: Shared Autonomy with PyTorch (Physical AI)

Session: "Shared Autonomy with PyTorch: Building Intelligent Physical AI Systems." This one leaves the chatbot world for robots.

E1. What PyTorch Is

Simple definition: PyTorch is the world's most popular open-source library for building and training neural networks. If Python is the language of AI, PyTorch is its most-used dialect — the tool nearly every model (LLMs, vision, robotics) is built in. Originally from Meta, now community-governed.

Telugu: PyTorch ante neural networks build + train cheyyadaniki prapancham lo most popular library (Python lo). Almost anni AI models — LLMs, vision, robots — PyTorch lo ne kadataru. AMD daaniki mukhyam endukante ROCm meeda PyTorch run avtund ani prove cheyyali.

Why it appears at an AMD event: PyTorch running well on ROCm is *the* proof that "AMD can do real AI." Every framework demo silently rests on PyTorch-on-ROCm.

E2. Physical AI & "Shared Autonomy"

Simple definition: **Physical AI** = AI that senses and acts in the *real world* (robots, drones, self-driving), not just text on a screen. **Shared autonomy** = human and machine *share control* — the AI handles the fine, fast, or dangerous parts while the human keeps intent and final say.

Telugu: Physical AI ante real world lo pani chese AI — robots, drones, self-driving cars (screen meeda text kaadu). Shared autonomy ante human + machine kalisi control cheyyatam — AI chinna/fast/dangerous panulu chestundi, human intent + final decision pettukumtaadu. Example: surgeon robot ni guide chestaadu, robot chetulu steady ga pedutumdi.

Analogy: Modern car lane-assist. You steer (intent); the car nudges you straight and brakes if you miss danger (fast correction). Neither fully drives — you *share* it. That's shared autonomy.

E3. Real-World Scenarios

- **Assistive robotics:** a wheelchair-mounted arm where the user says "pick that up" and the AI handles the precise grasp.
- **Surgical robots:** the surgeon directs; the system filters hand tremor and enforces safe boundaries.
- **Teleoperation + autonomy:** a human oversees a warehouse robot fleet; AI drives routine moves, human handles exceptions.
- **Advanced driver assistance (ADAS):** human + car sharing the driving task.

E4. Mini-Project: A Tiny PyTorch Control Policy

Goal: feel PyTorch + "an agent learning to control something" without any robot. ~1 hour, free. You'll train a tiny neural net to balance a pole (the "hello world" of control AI).

Step 1 — Install.

```
!pip install torch gymnasium
```

Step 2 — Meet the environment. `CartPole` = a cart that must move left/right to keep a pole upright. It's the classic control task — a stand-in for "robot keeping balance."

```
import gymnasium as gym
env = gym.make("CartPole-v1")
obs, _ = env.reset()
print(obs)    # 4 numbers: cart position, velocity, pole angle, angular velocity
```

Step 3 — Build a 2-layer policy net in PyTorch. Input the 4 sensor numbers → output "push left or right."

```
import torch, torch.nn as nn
policy = nn.Sequential(
    nn.Linear(4, 64), nn.ReLU(),
    nn.Linear(64, 2)      # 2 actions: left / right
)
```

Step 4 — Train it (reinforcement learning again!). Reward = +1 per timestep the pole stays up; the net adjusts to earn more. (Full training loop is ~30 lines — plenty of copy-paste notebooks exist; the point is to *run* one.)

Step 5 — Watch it learn to balance. Early on the pole falls instantly; after training it holds for hundreds of steps. You just built a *policy* — the exact concept behind "shared autonomy" robot control, minus the hardware.

Connect the dots: this is the *same* RL idea as Part D — a policy optimizing a reward — but the "environment" is physics, not text. That's why the event pairs them.

Part F — Concept: Model Serving & the OpenClaw Agent

Workshop (Track A & B): "Build Your OpenClaw Agent with Multi-Modal Models" — open-source multi-modal models, vLLM/SGLang serving. Two big ideas: serving and agents.

F1. Inference Serving — Why vLLM & SGLang Exist

Simple definition: *Training* makes a model; *inference* is using it to answer. **Serving** is running a model efficiently so *many* users get fast answers at once. **vLLM** and **SGLang** are the two leading open-source *inference servers* that make LLMs fast and cheap to run at scale.

Telugu: Training = model tayararu cheyyatam. Inference = daanini vadi javabu teppinchatam. Serving = chala mandiki okesari fast javabulu icchela model ni efficient ga run cheyyatam. vLLM, SGLang ane rendu open-source servers ee pani chestayi — LLM ni fast + cheap ga run chestayi.

Why they're needed (the KV-cache problem): naively, an LLM re-reads the whole conversation for every new token — hugely wasteful. vLLM's famous trick, **PagedAttention**, manages the model's short-term memory (the "KV cache") like an OS manages RAM pages — packing far more users onto one GPU. **Continuous batching** keeps the GPU busy instead of waiting. Result: often 5–20× more throughput than naive serving.

Tool	One-liner
vLLM	The throughput king; PagedAttention + continuous batching; OpenAI-compatible API
SGLang	Newer; great for <i>structured</i> + multi-step + agent workloads; fast growing

Both run on ROCm — so you can serve models on AMD GPUs. That's the workshop's foundation.

F2. Multi-Modal Models

Simple definition: A multi-modal model handles more than text — it can *see* (images), and sometimes hear or watch (audio/video), plus read and write text. "Describe this photo," "read this screenshot," "what's in this chart" — all multi-modal.

Telugu: Multi-modal model ante sirf text kaadu — images kuda chudagaladu (audio/video kudaa konni). "Ee photo lo enti?", "ee screenshot chaduvu" — ivi multi-modal panulu. Agent ki kallu vachchinattu.

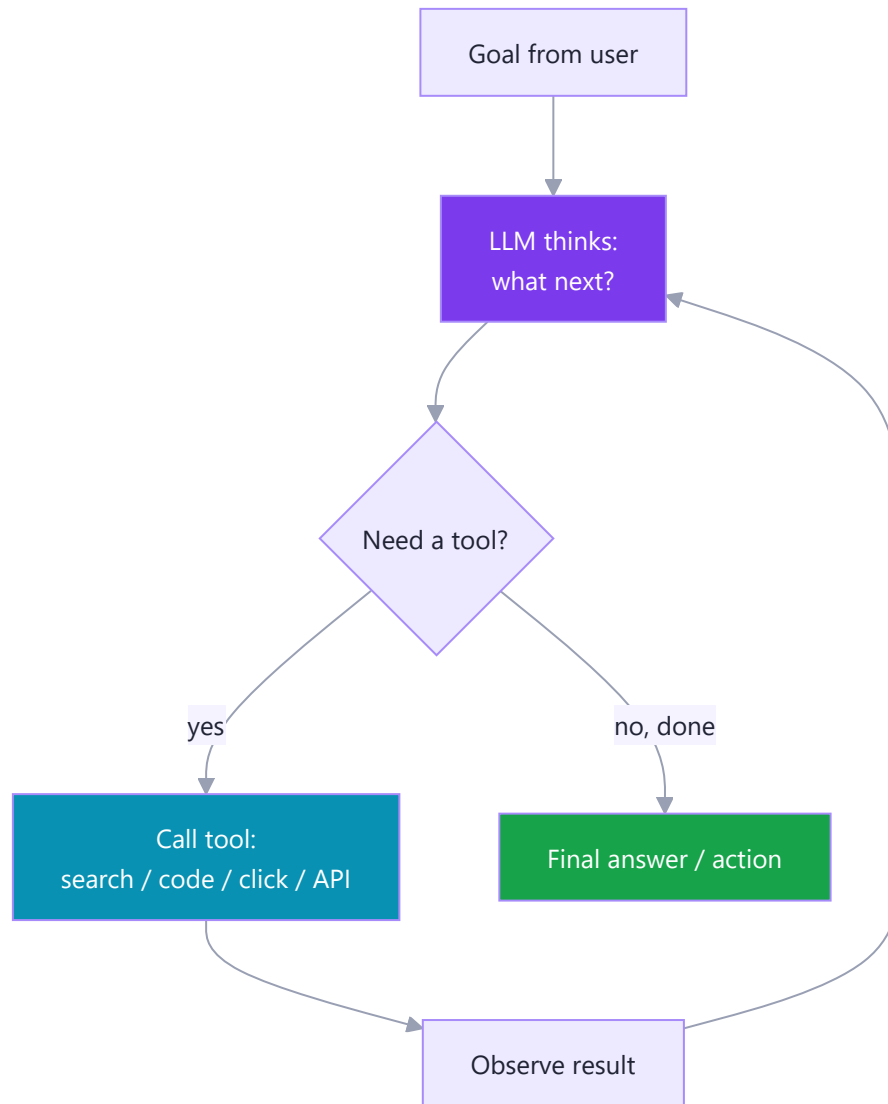
For an agent, multi-modal = the agent gets *eyes*. It can look at a webpage screenshot, a diagram, or a product photo and act on it — far more capable than a text-only agent.

F3. What an Agent Actually Is (OpenClaw)

Simple definition: An AI **agent** is an LLM that runs in a *loop* and can *use tools* to accomplish a goal — not just answer once. It thinks → picks a tool (search, run code, click, call an API) → sees the result → thinks again → repeats until done. **OpenClaw** is an open-source agent you'll build in the workshop.

Telugu: Agent ante LLM ni oka loop lo petti, tools vadadaniki icchi, oka goal complete cheyyinchatam. Alochinchu → tool vaadu (search, code run, click) → result chudu → malli alochinchu → goal ayyevaraku repeat. Meeru already Phase 12 lo Claude agents chesaaru — same idea, ippudu open models + AMD hardware meeda.

You already know this from your roadmap's agentic AI phase (Claude API + tool use + the agent loop). OpenClaw is the same pattern with *open* models served locally/on-AMD instead of a cloud API.



F4. Real-World Agent Scenarios

- **Coding agent:** "fix this failing test" → reads files, edits code, runs tests, iterates (your Claude Code experience).
- **Research agent:** "compare these 3 GPUs" → searches, opens pages, extracts specs, writes a table.
- **Computer-use agent:** looks at the screen (multi-modal), clicks buttons, fills forms.
- **Enterprise agent:** "reconcile these invoices" → reads PDFs, queries a DB, flags mismatches (ties straight to your SAP path).

F5. Mini-Project: Serve a Model + a Tool-Using Agent

Goal: serve an open model and point a tiny agent at it — the exact shape of the OpenClaw workshop. ~1–2 hours.

Step 1 — Serve a model (two easy paths).

- *No-GPU path (works on any laptop today):* install **Ollama** (ollama.com) → `ollama run qwen2.5:3b`. It exposes an **OpenAI-compatible** API at `localhost:11434` — the *same shape* as vLLM/SGLang/Lemonade, just simpler. Perfect for learning the pattern now.
- *GPU/cloud path (closer to the workshop):* `pip install vllm` then `vllm serve Qwen/Qwen2.5-3B-Instruct` on a Colab/rented GPU.

Step 2 — Talk to it with the OpenAI SDK. Because it's OpenAI-compatible, your code barely changes between local and cloud:

```
from openai import OpenAI
client = OpenAI(base_url="http://localhost:11434/v1", api_key="none") # local!
r = client.chat.completions.create(
    model="qwen2.5:3b",
    messages=[{"role": "user", "content": "In one line: what is an AI agent?"}],
)
print(r.choices[0].message.content)
```

Step 3 — Give it a tool (make it an agent). Define a Python function (e.g. a calculator or a web search), describe it to the model, and loop: if the model asks to call the tool, run it, feed the result back, repeat. That loop *is* the agent. (Frameworks like **LangChain**, **LlamaIndex**, or plain function-calling all do this — even ~40 lines of your own, like your Phase 12 `runAgent`.)

Step 4 — Add eyes (optional, multi-modal). Swap in a vision model (e.g. a Qwen-VL) and pass an image URL; ask "what's in this screenshot?" Now your agent is multi-modal — exactly OpenClaw's premise.

What to bring to the workshop: a working Ollama + a 40-line agent loop already on your laptop. You'll then swap Ollama → Lemonade/vLLM and your open model onto AMD hardware and *understand every step* instead of copy-pasting.

Part G — Concept: Vibe Coding with a Local Model

Workshop (Track B): "Vibe Coding with Local Model" — on-device AI coding on Ryzen AI Max 395 using Lemonade Server, Qwen Coder, and specialized AI agents. This is the flagship "AMD local AI" demo.

G1. "Vibe Coding" Defined

Simple definition: "Vibe coding" (term popularized by Andrej Karpathy) means building software by *describing what you want in natural language* and letting an AI write most of the code — you steer, review, and refine, working with the "vibe" of the idea rather than typing every line. Here, the twist: the AI runs **entirely on your laptop**, no cloud.

Telugu: Vibe coding ante — code okko line type cheyyakumdaa, "naaku idi kavali" ani English lo cheppi, AI tho code rayainchatam. Meeru steer chestaaru, review chestaaru. Ee session lo special enti ante — AI motham mee laptop lo ne run avtundi (Ryzen AI Max), cloud avasaram ledu. Private + offline + free.

Reality check (say this to sound senior): vibe coding is great for prototypes, demos, and glue code; production still needs you to read, test, and understand what the AI wrote. It's a *power tool*, not autopilot.

G2. Local vs Cloud Models — The Trade-Offs

	Cloud (ChatGPT, Claude)	Local (Lemonade + Ryzen AI)
Privacy	Data leaves your machine	100% on-device, private
Cost	Per-token, forever	Free after hardware
Offline	No	Yes
Model size	Huge (frontier)	Limited by your memory
Speed	Very fast (big servers)	Good, hardware-dependent
Best for	Hardest tasks, latest models	Private code, offline, cost-sensitive, high-volume

Telugu: Cloud model — pedddi, fast, kaani data byaataki potundi + prati token ki dabbu. Local model — private, offline, free (hardware konnaka), kaani size limited. Company sensitive code, offline work, or ekkuva volume unte local better. Strix Halo 128GB memory valla pedda local models kuda run avtayi.

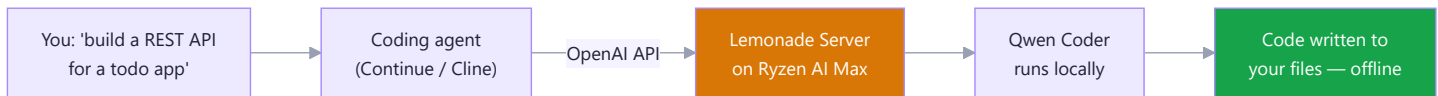
Why AMD wins this slide: local AI is memory-bound (Part B5). Strix Halo's 128GB unified memory lets it run *big* local models that a normal laptop's 8–16GB GPU can't — AMD's genuine edge over both NVIDIA laptops and Apple at the high end.

G3. Lemonade + Qwen Coder on Ryzen AI Max 395

The exact stack in the workshop:

- Ryzen AI Max+ 395 (Strix Halo)** — the hardware (Part C5): CPU + Radeon iGPU + XDNA NPU + 128GB unified memory.

- **Lemonade Server** (Part C6) — runs the model on NPU/iGPU, exposes an OpenAI-compatible API locally.
- **Qwen Coder** — Alibaba's open, code-specialized LLM family (excellent, permissive license) — the "brain."
- **Specialized agents** — coding agents (like Continue, Cline, or Aider) that plug into the local server and edit your codebase.



G4. Real-World Scenarios

- **Regulated industries** (banking, health, defense) where code/data *cannot* leave the building — local models are the only option.
- **Flights / remote sites** — coding assistant with no internet.
- **Cost at scale** — a team hammering an AI all day pays nothing per token locally.
- **Your SAP future** — enterprises increasingly want on-prem AI for the same privacy reasons; this skill transfers.

G5. Mini-Project: A Local Coding Agent

Goal: run a local coding assistant on your own laptop today — even without a Ryzen AI Max — so the workshop is "faster version of what I did" not "magic." ~1 hour.

Step 1 — Get a local server running. Easiest: **Ollama** (`ollama run qwen2.5-coder:7b`) — or, if you have an AMD Ryzen AI machine, install **Lemonade** (lemonade-server.ai / AMD's GitHub) to use the NPU. Both expose an OpenAI-compatible endpoint.

Step 2 — Install a coding agent in VS Code. Add the **Continue** or **Cline** extension. In its settings, point the model to your local server (base URL `http://localhost:11434/v1` , model `qwen2.5-coder:7b`).

Step 3 — Vibe-code something small. Open an empty folder and prompt: "Create an Express API with GET/POST /todos in memory, plus a test." Watch the local model write files. Review, run, fix — that's the loop.

Step 4 — Feel the difference. Note the speed, the privacy (airplane-mode it — still works!), and the size ceiling (a 7B model is good, not GPT-5). Now you *understand* exactly what Strix Halo's 128GB buys: bigger, smarter local models.

You're a web dev — this is the single most directly useful session for your day job. Come with Continue/Cline already working locally.

Part H — Concept: Fine-Tuning with Unsloth

Workshop (Track A): "Unsloth Fine Tuning LLM" — model customization on AMD Radeon GPUs with Unsloth Studio.

H1. Fine-Tuning vs Prompting vs RAG

Three ways to make a model do *your* task. Knowing when to use which is a senior-level distinction.

Approach	What it is	Best when	Cost
Prompting	Just write better instructions/examples	The base model already can do it	Free, instant
RAG	Retrieve your documents, stuff them into the prompt	The model needs your <i>facts/knowledge</i>	Cheap, no training
Fine-tuning	Actually adjust the model's weights on your data	You need a new <i>skill/style/format</i> , or smaller-model efficiency	Needs GPUs + data

Telugu: Model ni mee pani ki marchadaniki 3 daarulu. Prompting — better instructions ivvatam (free). RAG — mee documents ni prompt lo joddi facts ivvatam (cheap, training ledu). Fine-tuning — model weights ne mee data meeda train cheyyatam (kotta skill/style kavalmta). Prompting/RAG saripodu anappude fine-tune cheyyali.

The trap: beginners fine-tune when they should just RAG or prompt. Fine-tune for *behavior/skill/format* (e.g., "always answer in our brand voice as JSON"), not for *knowledge* (use RAG for that).

H2. LoRA & QLoRA — Cheap Fine-Tuning

Simple definition: Full fine-tuning updates all billions of weights — hugely expensive. **LoRA** (Low-Rank Adaptation) freezes the model and trains a *tiny* set of add-on weights (often <1% of the model) — 100× cheaper, nearly as good. **QLoRA** adds 4-bit quantization so it fits on a *single consumer GPU*.

Telugu: Full fine-tune ante anni billion weights marchatam — chala kharidu. LoRA — model ni freeze chesi, chinna add-on weights (1% kanna takkuva) matrame train cheyyatam — 100x cheap, almost same quality. QLoRA — daaniki 4-bit quantization joddm, so okka normal GPU (Radeon) meeda ne fine-tune cheyyochu. Anduke inti daggara possible.

Analogy: Instead of repainting a whole house (full fine-tune), you add removable sticky notes with corrections (LoRA adapters). Cheap, reversible, and you can keep many "note sets" for different tasks.

H3. What Unsloth Does

Simple definition: **Unsloth** is an open-source library that makes LoRA/QLoRA fine-tuning **~2× faster and ~50–70% less memory** through hand-optimized GPU kernels — so you can fine-tune real models on modest hardware (a single Radeon or a free Colab GPU). **Unsloth Studio** is its friendlier UI.

Telugu: Unsloth ante fine-tuning ni 2x fast + sagam memory tho cheyyinche library — special ga raasina GPU kernels valla. So okka Radeon GPU meeda kuda real models fine-tune cheyyochu. Ee workshop lo AMD Radeon + Unsloth Studio vaadataru — "inti GPU tho fine-tune cheyyochu" ane message.

Why AMD showcases it: Unsloth on Radeon = "you don't need an NVIDIA data center to customize a model." Democratizing fine-tuning is the pitch.

H4. Real-World Scenarios

- **Brand voice / format:** a support bot that always replies in your company's tone and a fixed JSON schema.
- **Domain jargon:** a legal/medical/SAP-ABAP model fluent in your niche vocabulary.
- **Smaller-cheaper:** fine-tune a 3B model to match a 70B model *on your narrow task* — huge cost savings at inference.
- **Style transfer:** make a model write in a specific author's or teacher's style.

H5. Mini-Project: Fine-Tune a Small Model

Goal: complete one real LoRA fine-tune end-to-end on free Colab, so the workshop is revision. ~1 hour.

Step 1 — Open Unsloth's official Colab. Unsloth publishes ready notebooks (search "Unsloth Colab Llama/Qwen"). Runtime → GPU.

Step 2 — Install + load a 4-bit model.

```
!pip install unsloth
from unsloth import FastLanguageModel
model, tokenizer = FastLanguageModel.from_pretrained(
    "unsloth/Qwen2.5-3B-Instruct-bnb-4bit",    # 4-bit = fits free GPU
    max_seq_length=2048, load_in_4bit=True,
)
model = FastLanguageModel.get_peft_model(model, r=16)    # add LoRA adapters
```

Step 3 — Bring a tiny dataset. 50–200 examples of "input → your desired output" (e.g., "customer message → polite JSON reply"). Format as chat pairs.

Step 4 — Train. A LoRA run of a few hundred steps takes minutes on the free GPU. Watch loss drop.

Step 5 — Test + save. Prompt the model with a *new* input; see it produce your target style. Save the small LoRA adapter file (a few MB — that's the "sticky notes," not the whole model).

Step 6 — Reflect. You changed a model's *behavior* for <₹0. At the workshop, the same runs on a Radeon GPU via ROCm — note the speed and that it's the *identical* code path.

What to ask: *"For a 3B model on a single Radeon, what dataset size and rank (r) hits the sweet spot before overfitting?"*

Part I — Concept: Video Generation with ComfyUI

Workshop (Track A & B): "Insider Look Into Captivating Video Generation (ComfyUI)" — prompt-based video workflows on PyTorch + GPU compute.

11. Diffusion Models — How AI Paints

Simple definition: Image/video generators use **diffusion**: they learn to turn pure noise into a picture, step by step. Training teaches the model to *remove* a little noise at a time; generation runs that in reverse — start from random static, denoise ~20–50 steps guided by your text prompt, and an image emerges. Video = do this for many frames, kept consistent over time.

Telugu: Image/video AI "diffusion" vaadutundi. Random noise (TV static la) tho modalu petti, mee prompt prakaaram konchem konchem noise teesi, 20-50 steps lo image ni byataiki testundi. Video ante ide chala frames ki chesi, time paatu consistent ga unchatam. Anduke video generation GPU-heavy — chala steps × chala frames.

Analogy: A sculptor starting with a rough marble block (noise) and chipping away (denoising) guided by a description until a statue (image) appears. Popular models: Stable Diffusion (image), and video models like Stable Video Diffusion / Wan / LTX.

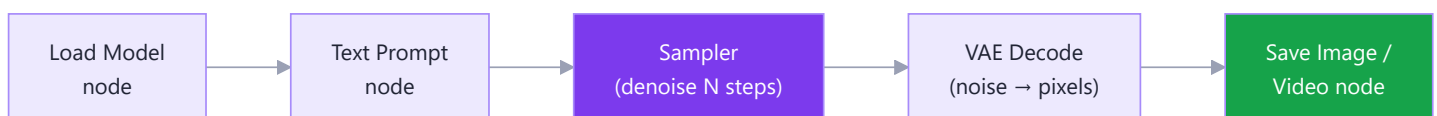
Why it's GPU-heavy: video = (many denoising steps) × (many frames) × (high resolution). This is where GPU memory and compute — AMD Radeon/Instinct — matter, and why it's an AMD showcase.

12. What ComfyUI Is (the Node Graph)

Simple definition: **ComfyUI** is a free, open-source, *node-based* interface for building image/video generation pipelines visually — you drag boxes (load model → set prompt → sample → decode → save) and wire them together, instead of writing code. It's the power-user tool for generative media (runs on PyTorch, supports AMD via ROCm).

Telugu: ComfyUI ante image/video generate cheyyadaniki node-based visual tool. Code rayaakumdaa boxes (model load → prompt → sample → decode → save) drag chesi wire chestaaru. n8n workflow builder gurtundaa? Same feel — kaani AI images/videos kosam. AMD GPUs meeda ROCm tho run avtund.

You know this feeling — it's like the n8n visual workflows from your roadmap's Phase 12, but the nodes are generation steps. If you can wire an n8n flow, you can read a ComfyUI graph.



13. Text-to-Video vs Image-to-Video

Mode	Input → Output	Use
Text-to-Video	Prompt → clip	"A drone shot over Bengaluru at sunset"
Image-to-Video	Still image → animated clip	Bring a product photo or logo to life
Video-to-Video	Clip → restyled clip	Change style/season, apply an effect

Telugu: Text-to-video — matalu ichchī clip. Image-to-video — oka photo animate cheyyatam. Video-to-video — unna clip ni style marchadam. Workshop lo prompt ichchī clip generate cheyyadam nerpistaaru.

14. Real-World Scenarios

- **Marketing/social:** generate product teasers and ads without a film crew.
- **Indie creators:** animate concept art, make music-video visuals.
- **Prototyping:** storyboard a scene in minutes.
- **Localization:** regenerate a clip's style/text per region.

15. Mini-Project: First ComfyUI Video Workflow

Goal: generate one image, then one short clip, in ComfyUI — so the workshop's graph is familiar. ~1–2 hours (GPU helps; Colab/cloud works).

Step 1 — Get ComfyUI. Install locally (has an **AMD/ROCm** path — see ComfyUI docs) or use a hosted ComfyUI (e.g. a Colab notebook or a cloud GPU service) if your machine lacks a strong GPU.

Step 2 — Run the default image graph. ComfyUI ships a text-to-image workflow. Load a Stable Diffusion checkpoint, type a prompt, hit **Queue Prompt**, watch the nodes light up and an image appear. *Read each node* — that's the diffusion pipeline from I1 made visual.

Step 3 — Add image-to-video. Load a video model (e.g. Stable Video Diffusion) via a template workflow; feed your generated image in; produce a 2–4 second clip. Expect it to be slow — that's the compute cost being real.

Step 4 — Tweak one knob at a time. Change steps, seed, prompt, motion strength — see the effect. This builds intuition for what the workshop presenter is doing live.

Step 5 — Note the hardware reality. Watch VRAM usage; hit limits; understand why "more GPU memory" (AMD's pitch) directly means "longer/higher-res video."

What to ask: "On Radeon via ROCm, which video models run well today, and where's the VRAM ceiling for 720p clips?"

Part J — Concept: Sovereign AI & the India AI Ecosystem

Sessions: "Sovereign India AI Eco System Gen" (partner) and "From Campus to Code" (AMD University). Less code, more strategy — but it's why the event is in India.

J1. What "Sovereign AI" Means

Simple definition: Sovereign AI is a country building and running its *own* AI — models, data, and the compute (chips/data centers) — inside its borders, under its own control and languages, rather than depending entirely on foreign (mostly US) providers.

Telugu: Sovereign AI ante oka desam sonta AI build chesukovadam — models, data, computers anni tama desam lopala, tama control lo, tama bhashallo. Vere desam (mostly US) meeda purtiga adharpadakumdaa. India ki idi mukhyam — Hindi, Telugu, Tamil laanti bhashallo, Indian data tho AI kavali.

Why it matters: data privacy, national security, cost, and — crucially for India — **language and culture**. A model trained mostly on English/US data serves Indian users poorly. Sovereign AI = models fluent in Indian languages, aware of Indian context.

J2. India's AI Push

- **IndiaAI Mission** — a national program funding shared GPU compute, datasets, and startups.
- **Indic LLMs** — models built for Indian languages (e.g. efforts like Sarvam, Krutrim, BharatGPT, and academic groups).
- **Shared compute** — subsidized GPU access so startups/researchers aren't priced out.

Telugu: India lo "IndiaAI Mission" — GPU compute, datasets, startups ki funding icche national program. Indian bhashala kosam models (Sarovam, Krutrim laantivi). Shared GPU access — chinna companies/researchers ki cheap compute. AMD ee ecosystem lo chip supplier avvali anukuntundi.

J3. Why AMD Cares

Sovereign AI = countries buying **lots of GPUs and building data centers**. AMD wants to be that supplier (an alternative to NVIDIA) for India's sovereign compute — hence a *partner* talk on the India AI ecosystem. For AMD, "sovereign AI" is both a mission and a massive market.

J4. From Campus to Code

AMD University grows the developer pipeline — students learning ROCm/Ryzen AI today become the engineers who *choose* AMD tomorrow. This session (and the awards) is AMD investing in India's student/developer community. As a career-switcher, this is your networking gold: mentors, hiring managers, and a community betting on the same open-AI future.

Telugu: "From Campus to Code" — students ki AMD tech nerpinchi, repu vaalle AMD choose chesela cheyyatam. Meeku idi networking ki best — mentors, hiring managers akkada untaaru. Career switch chestunna meeku ee connections chala viluvainavi.

Part K — Your Step-by-Step Prep Plan

You have ~12 days (today ≈ Jul 27 → event Aug 8). Here's exactly how to spend the mornings so you walk in ready — not overwhelmed. ~1 hour/day is enough; do the projects on weekends.

K1. The 12-Day Countdown

Day	Focus	Concrete action
1	AMD story	Read Part A. Watch a "Lisa Su AMD turnaround" video + "What is ROCm" (10 min each)
2	Foundations	Read Part B. Open Colab, run 10 Python cells (Part B1)
3	Training pipeline	Re-read B3 + Part D. Watch "RLHF explained" + "GRPO/DeepSeek" (short videos)
4	RL project	Do the TRL mini-project (D5) on Colab
5	AMD stack	Read Part C. Watch "Ryzen AI Max 395 local LLM" + "Lemonade Server" demo
6	Serving + agents	Read Part F. Install Ollama , run <code>qwen2.5:3b</code> , hit it with the OpenAI SDK (F5 steps 1–2)
7	Agent project	Finish the agent loop (F5 steps 3–4) — reuse your Phase 12 <code>runAgent</code>
8	Vibe coding	Read Part G. Install Continue/Cline in VS Code → local model (G5). Vibe-code a small API
9	Fine-tuning	Read Part H. Do the Unsloth LoRA mini-project (H5) on Colab
10	Video gen	Read Part I. Run a ComfyUI text-to-image graph (I5 steps 1–2); read the nodes
11	Strategy + glossary	Read Parts J + L. Skim the glossary (L3). Pick your track (K3)
12	Light review	Re-skim your notes + the "one-liners." Pack (L1). Rest — you're ready

Telugu: 12 rojulu, roju ~1 gmta chaalu. Modati aaru rojulu concepts + AMD story + rendu chinna projects (RL, agent). Taruvata vibe coding, fine-tuning, video — okko roju okkati. Chivari roju rest. Weekends lo projects ki ekkuva time ivvu. Overwhelm avvakmdi — ee event beginners-friendly, meeru already agents (Phase 12) chesaaru.

K2. Accounts & Installs Checklist

Set these up *before* Aug 8 (venue Wi-Fi + install time will eat your workshop):

- [] **Google account** → Colab works (free GPU for the projects)
- [] **Hugging Face account** + an **access token** (models/datasets need it) — huggingface.co
- [] **Python 3.11+** installed locally + VS Code
- [] **Ollama** installed, one model pulled (`ollama run qwen2.5:3b`)
- [] **VS Code** + **Continue** (or **Cline**) extension
- [] **Git** + a **GitHub** account (to clone workshop repos)
- [] **Laptop fully charged + charger** (event explicitly says bring it)
- [] (Optional) a **cloud GPU** account (Colab Pro / Lightning / RunPod) as backup for heavier steps

- [] Downloaded a couple of small models *in advance* so you're not fighting venue Wi-Fi

K3. Track A vs Track B — Which to Pick

Sessions run in **parallel** — you can't do both, so decide now. (Note: OpenClaw and Video Generation each appear on both tracks at different times, so you can usually catch them regardless.)

If you care most about...	Lean	Because
Your web-dev day job / practical coding	Track B → Vibe Coding	Local coding agent is the most immediately useful skill
Becoming an AI/ML engineer	Track A → Unsloth Fine-Tuning	Customizing models is a core ML skill
Agents (your Phase 12 strength)	Either → OpenClaw	It's on both tracks; catch whichever slot
Creative / generative media	Either → Video Generation	Also on both tracks

Telugu: Rendu tracks okesari jarugutaayi — okkati matrame choodagalaru. Meeru web dev + practical ishtapadite → Track B (Vibe Coding). ML engineer avvaalante → Track A (Unsloth fine-tuning). OpenClaw agent + Video generation rendu tracks lo vstaayi, so avi miss avvāu. Munde decide chesukomdi, room lo time waste avvkumdaa.

My recommendation for you (web dev → SAP+AI): Track B / Vibe Coding. A private, local coding assistant is the skill you'll use next week — and it maps directly to enterprises wanting on-prem AI (your SAP future). Catch OpenClaw in its other-track slot for the agent content.

Part L — Day-Of Playbook & Glossary

L1. What to Bring & Set Up

- **Laptop + charger** (mandatory), fully charged.
- **All accounts + installs from K2 done** — do *not* rely on venue Wi-Fi for setup.
- **A HF token** saved somewhere handy.
- **This document** (offline copy) + a notebook for the speakers' repo links/QRs.
- **Business cards / LinkedIn ready** — networking (awards, "Campus to Code") is half the value.
- Arrive by **10:00–10:15 AM** for the kickoff; workshops fill (Track A 220, Track B 200 seats).

L2. Smart Questions to Ask Speakers

Asking one good question gets you remembered (and maybe hired). Steal these:

- **RL talk:** *"When you scale GRPO, is the bottleneck generation or the gradient step — and how does Instinct's 288GB memory change the trade-off?"*
- **Vibe Coding:** *"What's the biggest Qwen Coder model that runs comfortably on the 128GB Strix Halo, and at what tokens/sec?"*
- **Unsloth:** *"On a single Radeon via ROCm, what dataset size and LoRA rank avoid overfitting for a niche domain?"*
- **ComfyUI:** *"Which video models run well on Radeon today, and where's the VRAM ceiling for 720p?"*
- **Anyone:** *"Honestly, where does ROCm still hurt vs CUDA in your workflow?"* (Shows maturity; sparks a real answer.)

L3. Glossary — Quick Reference

Term	One-line meaning
CPU / GPU / NPU	Few smart cores / thousands of parallel cores / tiny power-efficient AI unit
ROCm	AMD's open-source CUDA rival — runs AI frameworks on AMD GPUs
CUDA	NVIDIA's proprietary GPU-compute layer (their historical moat)
HIP / HIPIFY	Write-once GPU code for both vendors / auto-convert CUDA→HIP
Instinct	AMD data-center AI GPUs (MI300X/MI350X)
Ryzen AI / XDNA	AMD laptop chips with an on-device AI NPU
Strix Halo (AI Max+ 395)	Top Ryzen AI chip; 128GB unified memory for big local models
Lemonade Server	AMD's local LLM server (OpenAI-compatible API)
GAIA	AMD's open-source local-agent framework
Token / Parameters	Word-piece the model reads-writes / the model's tunable knobs (7B = 7bn)
Quantization (4-bit)	Shrinking a model to fewer bits so it fits on smaller hardware
Context window	How many tokens the model can consider at once
Pretrain / SFT / RLHF	Learn language / learn to follow instructions / learn to prefer good answers
RLHF / Reward model	RL from human feedback / model that predicts human scores
PPO / GRPO / RLVR	Classic RL / cheaper newer RL / RL with test-based (verifiable) rewards
TRL	Hugging Face's RL fine-tuning library
LoRA / QLoRA	Cheap fine-tuning via tiny adapters / + 4-bit to fit one GPU
Unsloth	Library that makes LoRA/QLoRA ~2× faster, half the memory
vLLM / SGLang	High-throughput LLM inference servers (PagedAttention, batching)
Agent	LLM in a loop that uses tools to reach a goal
Multi-modal	Model that also handles images/audio/video, not just text
Diffusion	Noise→image generation method (Stable Diffusion, video models)
ComfyUI	Node-based visual tool for image/video generation
Vibe coding	Building software by describing intent; AI writes the code
RAG	Retrieve your docs into the prompt to give the model your facts
Sovereign AI	A country's own AI — models, data, compute, languages, in-borders

L4. After the Event

- **Star every GitHub repo** the speakers share (you'll forget names, not stars).
- **Redo one workshop** the same weekend while it's fresh — that's what makes it stick.
- **Post one takeaway on LinkedIn**, tag AMD/speakers — visibility for your career switch.
- **Fold it into your roadmap:** the local-model + agent skills slot directly into your Phase 12 (agentic AI) and your later SAP-on-prem-AI story.
- **Keep one project going:** pick the workshop you loved and build a small real thing with it over the next month.

The mindset: you're not going to master AMD's stack in one day — you're going to *demystify* it, meet the community, and leave with 3–4 concrete skills and projects. Come prepped with this doc, and every session upgrades from "interesting blur" to "oh, I've done a small version of this." That's the whole goal.

Part M — AMD Terms Glossary — One Para Each, with Years

The reference you revise on the metro to the venue: every AMD term from this doc — plus the silicon basics — one paragraph each, with the year it launched. Parts A and C tell the story; this part is the flash-card deck.

Telugu: Ee bhagam antha silicon gurinchi — nee software nadiche chips, mariyu AMD (nee DevDay company). Brackets lo unna years ante aa product release ayina year — ivi telisthe peru cheppadam kaadu, **artham chesukuni** matladagalav.

CPU (Central Processing Unit)

The CPU is the general-purpose brain: a few very powerful cores (4–16 in laptops, up to 192+ in servers) optimised for **low latency** — finishing *one* task as fast as possible, then the next. It runs your OS, your browser, your Node.js code — anything, in any order, with heavy branching logic. Its cores are complex: caches, branch predictors, out-of-order execution. **Example:** parsing JSON, running Express routes, executing SQL — serial, unpredictable, logic-heavy work is CPU territory. AMD's CPUs are Ryzen (consumer) and EPYC (server).

Telugu: CPU ante general-purpose **brain** — konni chala powerful cores, okka pani ni atyanta vega ga complete cheyyadaniki (low latency). OS, browser, nee Node code — edaina, e order lo aina nadipistundi. Logic ekkuvu unna pani antha CPU de.

GPU (Graphics Processing Unit)

The GPU is the parallel monster: **thousands** of small, simple cores that all do the same operation on different data simultaneously — built for graphics (millions of pixels, same maths each) and, it turned out, perfect for AI (neural networks are giant matrix multiplications). Optimised for **throughput**, not latency: any single operation is slower than a CPU's, but a million together finish far sooner. **Example:** training an LLM means multiplying billion-entry matrices — a 64-core CPU would take months; a rack of GPUs does it in days. AMD's GPUs: Radeon (gaming) and Instinct (AI/data centre).

Telugu: GPU ante **parallel monster** — velakoladi chinna cores, anni okate operation ni veru veru data meeda okesari chestayi. Graphics ki puttindi (lakshala pixels, same lekka), AI ki perfect ani telisindi (neural networks ante pedda matrix multiplication). Okka pani slow, kaani **lakshala panulu kalipi** chala fast.

NPU (Neural Processing Unit)

The NPU is the newest sibling (mainstream ~2023): a small accelerator dedicated to **AI inference at very low power** — running an already-trained model (transcription, background blur, local copilots) using a fraction of the battery a GPU would burn. It can't do graphics or general compute; it does one thing, efficiently, all day. Measured in **TOPS**. **Example:** live captions translating a video call on a laptop that stays cool and lasts the flight — that's the NPU working while CPU and GPU idle. AMD's NPU is the **XDNA** engine inside Ryzen AI chips (2023), descended from Xilinx technology.

Telugu: NPU ante kotha family member (~2023) — **chala takkuva power tho AI inference** matrame chese chinna chip. Trained model ni nadipistundi (transcription, background blur) — battery burn cheyakunda, rojantha. Graphics/general pani cheyyaledu — okate pani, efficient ga. AMD di: **XDNA** (Ryzen AI lo).

CPU vs GPU vs NPU — the one-table difference

Think of a restaurant: the CPU is the **head chef** — handles any dish, complex orders, decisions. The GPU is **500 kitchen helpers** all chopping onions at once — one repetitive task at massive scale. The NPU is the **rice cooker** — one job, silently, using almost no gas. Modern "AI PC" chips (like Ryzen AI) put all three on one die and route each task to whichever does it cheapest.

	CPU	GPU	NPU
Cores	Few, powerful	Thousands, simple	Specialised array
Optimised for	Latency (one task fast)	Throughput (many tasks at once)	Efficiency (AI per watt)
Best at	Logic, OS, apps, branching	Graphics, AI training , parallel maths	AI inference on-device
Power draw	Medium	Huge (300–700W+ for AI cards)	Tiny (single-digit watts)
Metric	GHz, cores	TFLOPS	TOPS
AMD product	Ryzen, EPYC	Radeon, Instinct	XDNA (Ryzen AI)

Telugu: Hotel analogy: CPU = **head chef** (edaina vandagalladu, decisions), GPU = **500 helpers** andaru okesari ullipayalu kosam (okate pani, massive scale), NPU = **rice cooker** (okate pani, gas daadapu zero). AI PC ante mudu okate chip lo — pani deniki cheap o daaniki veltundi.

AMD — Advanced Micro Devices (founded 1969)

AMD is one of the two companies (with Intel) that build the x86 processors most computers run on — and, since its 2010s comeback, the only company competing across **all four** silicon fronts: CPUs (Ryzen/EPYC), GPUs (Radeon/Instinct), NPUs (XDNA), and adaptive chips (Xilinx FPGAs). Founded in 1969 by Jerry Sanders; nearly bankrupt by 2014; revived by the Zen architecture into one of the world's most valuable chip companies. **Example:** the PS5, your gaming laptop, Frontier (the first exascale supercomputer, 2022), and a growing share of AI data centres all run on AMD silicon.

Telugu: AMD (1969, Jerry Sanders) ante Intel tho paatu x86 processors chese rendu companies lo okati — ippudu **naalugu fronts** lo pote unna okate company: CPU (Ryzen/EPYC), GPU (Radeon/Instinct), NPU (XDNA), adaptive chips (Xilinx). 2014 lo daadapu bankrupt; Zen tho tirigi lechi top chip companies lo okati ayyindi.

x86 & AMD64 (2003)

x86 is the instruction-set architecture — the vocabulary of instructions a CPU understands — that Intel created (1978) and AMD licenses and co-develops. AMD's historic move: in **2003** it invented the 64-bit extension (**AMD64 / x86-64**, launched with Opteron and Athlon 64) while Intel was betting on a different architecture — Intel ended up adopting *AMD's* design, and every Windows/Linux PC today runs it. **Example:** when you download

software labelled "x64" or "amd64" — including on an Intel machine — that name is AMD's 2003 victory fossilised into filenames.

Telugu: x86 ante CPU ki ardham ayye instructions bhasha (Intel, 1978). AMD 2003 lo daaniki **64-bit version (AMD64)** kanipettindi — Intel ye taruvata AMD design ni adopt chesindi. Ippatiki downloads lo kanipinche "amd64" aa gelupu gurthu — Intel machine meeda kuda.

Zen Architecture (2017)

Zen is the CPU core design that saved AMD: launched **March 2017** after years of falling behind, it delivered a ~52% performance-per-clock jump over AMD's previous cores and has iterated relentlessly — Zen 2 (2019, chiplets), Zen 3 (2020), Zen 4 (2022), Zen 5 (2024). Every Ryzen, Threadripper, and EPYC since 2017 is a Zen variant. **Example:** the comeback story interviewers and keynotes love: AMD stock was ~\$2 in 2015; Zen-based EPYC then took server market share from Intel every single quarter for years.

Telugu: Zen (March 2017) ante AMD ni kaapadina CPU core design — mundu generation kanna ~52% performance jump. Prathi Ryzen, EPYC deeni variant ye: Zen 2 (2019, chiplets), Zen 3 (2020), Zen 4 (2022), Zen 5 (2024). \$2 stock nunchi comeback ki punaadi ide.

Ryzen (2017)

Ryzen is AMD's consumer CPU brand — desktops and laptops — launched **March 2017** as the first Zen product, instantly making 8 cores mainstream when Intel was selling 4. Naming works like car tiers: Ryzen 3 (budget), 5 (mainstream), 7 (performance), 9 (enthusiast). **Example:** a "Ryzen 7 9800X3D" gaming build or the Ryzen inside most non-Apple laptops on sale today; the "X3D" variants stack extra cache in 3D — a chiplet-era trick.

Telugu: Ryzen (2017) ante AMD consumer CPU brand — desktops, laptops. Intel 4 cores ammutunte Ryzen 8 cores ni mainstream chesindi. Tiers car laaga: Ryzen 3 / 5 / 7 / 9.

Threadripper (2017)

Threadripper is Ryzen's extreme sibling (**August 2017**): workstation CPUs with enormous core counts (16 at launch; up to 96 today) for people whose work is parallel — 3D rendering, video, simulation, code compilation. It sits between desktop Ryzen and server EPYC. **Example:** a VFX artist rendering a film frame across 96 cores, or compiling a monster C++ codebase in minutes instead of an hour.

Telugu: Threadripper (2017) ante Ryzen ki extreme anna — 16 nunchi 96 cores workstation CPUs. Parallel pani chesevaallaki: 3D rendering, video, pedda code compile. Desktop Ryzen ki server EPYC ki madhyalo untundi.

EPYC (2017)

EPYC is AMD's server CPU line (**June 2017**) — the chips inside cloud data centres, with up to 192 cores per socket today. Its Zen-plus-chiplets design let AMD pack more cores per socket at better efficiency than Intel's Xeon, taking AMD's server share from ~1% (2017) to ~a third of the market. **Example:** many AWS instance types (the "a" in `m7a.large` means AMD) and the Frontier and El Capitan supercomputers run EPYC — your Phase 10 EC2 box may literally be one.

Telugu: EPYC (2017) ante AMD server CPU — cloud data centres lo undedi, socket ki 192 cores varaku. Intel Xeon kanna cores ekkuva, efficiency better — server market ~1% nunchi ~third ki perigindi. AWS lo [m7a](#) laanti "a" instances ante AMD.

APU & Semi-Custom (2011)

An APU (Accelerated Processing Unit) is AMD's term for CPU + GPU on one chip (first: Llano, **2011**) — one die does both compute and graphics, ideal for laptops and consoles. AMD's **semi-custom** business designs bespoke APUs for clients. **Example:** the PlayStation 4 and Xbox One (2013), PS5 and Xbox Series X (2020), and the Steam Deck (2022) are all AMD semi-custom APUs — practically every gaming console this decade is AMD inside.

Telugu: APU ante CPU + GPU okate chip lo (2011 nunchi) — laptops, consoles ki perfect. Semi-custom ante client kosam pratyekam ga design chesedi: PS4, PS5, Xbox, Steam Deck — ee decade lo daadapu prathi console lopala AMD ye.

Radeon (2000 · ATI acquired 2006)

Radeon is AMD's graphics brand for gamers and creators — born at ATI in **2000**, joining AMD when it bought ATI in **2006** (\$5.4B). It's the perennial rival to NVIDIA's GeForce; current cards are the RX series. **Example:** an "RX 9070 XT" in a gaming build, or the Radeon graphics integrated into every Ryzen laptop chip — plus, via semi-custom, the graphics inside your PS5.

Telugu: Radeon ante AMD gaming graphics brand — ATI lo puttindi (2000), 2006 lo AMD ATI ni konnappudu vachindi (\$5.4B). NVIDIA GeForce ki shaswatha rival. Prathi Ryzen laptop lo integrated Radeon untundi.

RDNA & CDNA (2019 · 2020)

In 2019 AMD split its GPU designs in two: **RDNA** (2019, RX 5000 series) optimised for gaming — frames per second, latency; **CDNA** (2020, Instinct MI100) optimised for compute — pure matrix throughput for AI and HPC, no display outputs at all. The 2025 roadmap merges lessons from both into **UDNA**. **Example:** RDNA renders Cyberpunk on your Radeon; CDNA multiplies matrices in a national lab — same DNA suffix, opposite jobs.

Telugu: 2019 lo AMD GPU designs rendu ga vidipoyayi: **RDNA** (2019) gaming ki — FPS, latency; **CDNA** (2020) compute ki — AI/HPC matrix maths matrame, display outputs kuda undavu. Same family, opposite panulu.

Instinct MI Series (2016 · MI300X 2023)

Instinct is AMD's data-centre AI accelerator line (first announced **2016**), and the **MI300X (December 2023)** is the one that made it real competition to NVIDIA — 192GB of memory per chip (vs H100's 80GB) meant bigger models fit on fewer GPUs. Successors: MI325X (2024), MI350 series (2025). **Example:** Microsoft Azure and Meta run MI300X fleets for LLM inference; when a keynote says "we serve GPT-scale models on AMD," this chip family is what they mean — expect it constantly at DevDay.

Telugu: Instinct ante AMD data-centre AI accelerator (2016 nunchi); **MI300X (Dec 2023)** tho NVIDIA ki nijamaina poti modalayyindi — chip ki 192GB memory (H100 di 80GB), pedda models takkuva GPUs meeda pattestayi. Microsoft, Meta LLM inference ki vaadutunnayi. DevDay lo idi malli malli vintav.

ROCm (2016)

ROCm (Radeon Open Compute) is AMD's open-source software stack for GPU computing — its answer to NVIDIA's CUDA, launched **2016**. It's the layer that lets PyTorch and TensorFlow run on AMD GPUs; its HIP tool converts CUDA code to portable code. NVIDIA's real moat is CUDA's 15-year head start, so ROCm's maturity is *the* strategic question in AI hardware. **Example:** `pip install torch` with the ROCm build, and your PyTorch model trains on an MI300X with no code changes — that sentence becoming routinely true is what AMD's AI future depends on.

Telugu: ROCm (2016) ante AMD open-source GPU software stack — NVIDIA CUDA ki jawab. PyTorch/TensorFlow ni AMD GPUs meeda nadipinchedi ide. NVIDIA asli balam CUDA ki unna 15 yella head start — anduke ROCm maturity ye AI hardware lo pedda strategic prashna.

Ryzen AI & XDNA (2023)

Ryzen AI is AMD's brand for laptop chips with a built-in NPU, and XDNA is that NPU's architecture — technology inherited from the Xilinx acquisition, first shipped in the Ryzen 7040 series (**2023**), the first x86 laptop chips with an NPU. XDNA 2 (**2024**, Ryzen AI 300) reaches ~50 TOPS, clearing Microsoft's 40-TOPS bar for **Copilot+ PC** features. **Example:** a Ryzen AI laptop running local Windows Studio Effects and small LLMs offline, on battery — the "AI PC" category every chip company is now chasing.

Telugu: Ryzen AI ante NPU unna laptop chips brand; **XDNA** aa NPU architecture — Xilinx nunchi vachina technology, 2023 lo Ryzen 7040 tho first x86 laptop NPU. XDNA 2 (2024) ~50 TOPS — Microsoft Copilot+ PC bar (40 TOPS) daati. 'AI PC' race idi.

Infinity Fabric & Chiplets (2017 · 2019)

Infinity Fabric (**2017**) is AMD's high-speed interconnect — the nervous system linking cores, memory, and chips. It enabled the **chiplet** revolution (mainstream with Zen 2, **2019**): instead of one huge expensive die, build several small dies and wire them together — small dies have far better manufacturing yield, so more cores for less money. The whole industry followed. **Example:** a 96-core EPYC is ~13 chiplets on one package; the MI300X mixes CPU, GPU, and memory chiplets in 3D — impossible as a single die.

Telugu: Infinity Fabric (2017) ante chips lopala/madhya high-speed connection — nervous system. Adi **chiplets** (2019, Zen 2) ni saadhyam chesindi: okka pedda die badulu chinna dies katti kaluputaru — yield better, cores ekkuva, khreedu takkuva. Industry antha follow ayyindi.

Xilinx & the FPGA (1985 · acquired 2022)

Xilinx invented the **FPGA** (first chip **1985**) — a Field-Programmable Gate Array, a chip whose *circuits themselves* can be rewired by software after manufacturing: hardware you can reprogram like code. AMD bought Xilinx in **February 2022** for ~\$49B (then the largest semiconductor acquisition ever), gaining the adaptive-computing portfolio and the AI-engine technology that became XDNA. **Example:** 5G base stations, Mars rovers, and stock-exchange trading systems use FPGAs where algorithms must change after deployment but CPUs are too slow — telecom gear updates its silicon behaviour via a file.

Telugu: Xilinx **FPGA** ni kanipettindi (1985) — manufacture ayyaka kuda **circuits ni software tho re-wire** cheyagalige chip; code laaga reprogram chese hardware. AMD 2022 lo ~\$49B ki konnadi (appatlo semiconductor history lo largest deal). XDNA NPU technology ikkadinunchey vachindi.

Versal & Alveo (2018)

Two flagship Xilinx product lines now under AMD. **Versal** (announced **2018**) is an "adaptive SoC": FPGA fabric + ARM CPU cores + AI engines on one chip — for 5G, automotive, aerospace. **Alveo (2018)** packages FPGA power as plug-in data-centre accelerator cards for workloads like video transcoding and network processing. **Example:** a broadcaster transcoding hundreds of live streams on Alveo cards, or a car's driver-assist system running on a Versal chip that gets smarter with each over-the-air update.

Telugu: Rendu Xilinx flagship lines: **Versal** (2018) = FPGA + ARM CPU + AI engines okate chip lo — 5G, cars, aerospace ki. **Alveo** (2018) = data centre lo pettukune FPGA accelerator cards — video transcoding laanti panulaki.

Pensando & the DPU (acquired 2022)

Pensando (bought by AMD, **May 2022**, ~\$1.9B) makes **DPUs** — Data Processing Units: smart network chips that offload networking, storage, and security work from the CPU, so every server core does *paying* work instead of plumbing. The trio "CPU + GPU + DPU" is the modern data-centre formula. **Example:** in a cloud data centre, encryption, firewalling, and traffic routing run on the DPU in the network card — freeing whole CPU cores per server, which at 100,000 servers is a fortune.

Telugu: Pensando (AMD konnadi 2022, ~\$1.9B) **DPU** lu chestundi — networking, storage, security pani ni CPU nunchi teesukune smart network chips. Server cores anni **dabbulu vacche pani** ke ankitham — plumbing DPU chusukuntundi. Modern data centre formula: CPU + GPU + DPU.

ZT Systems (acquired 2025)

ZT Systems builds **rack-scale AI infrastructure** — not chips but the complete racks: thousands of GPUs integrated with cooling, power, and networking, ready to roll into a data centre. AMD bought it (announced **August 2024**, completed **March 2025**, \$4.9B) to sell what NVIDIA sells: full AI *systems*, not loose chips — keeping ZT's ~1,000 design engineers while selling the manufacturing arm (to Sanmina, 2025). **Example:** a cloud provider ordering "an AI cluster" now gets AMD-designed complete racks of MI-series GPUs — the acquisition is why that sentence is possible, and why you'll hear "rack-scale" at DevDay.

Telugu: ZT Systems chips kaadu — **complete AI racks** kadutundi: veyyi GPUs + cooling + power + networking, data centre lo pettadame taruvayi. AMD \$4.9B ki konnadi (complete March 2025) — NVIDIA laaga **full systems** ammadaniki, chips matrame kaadu. DevDay lo 'rack-scale' ane maata deeni valle.

TOPS (Tera Operations Per Second)

TOPS is the marketing-and-engineering unit for NPU speed: trillions of (usually low-precision INT8) operations per second. It's how "AI PC" capability is graded — but like megapixels, more TOPS doesn't automatically mean better real-world AI; memory and software matter too. **Example:** Microsoft's Copilot+ PC badge requires **40+ TOPS**; XDNA 2 delivers ~50. When a spec sheet shouts "80 TOTAL AI TOPS," read the footnote — that often sums CPU + GPU + NPU.

Telugu: TOPS ante NPU speed unit — second ki trillions of operations. 'AI PC' grade cheyyadaniki idi vaadutaru: Copilot+ badge ki **40+ TOPS** kaavali, XDNA 2 ~50 istundi. Kaani megapixels laage — TOPS ekkuva ante automatic ga AI better ani kaadu.

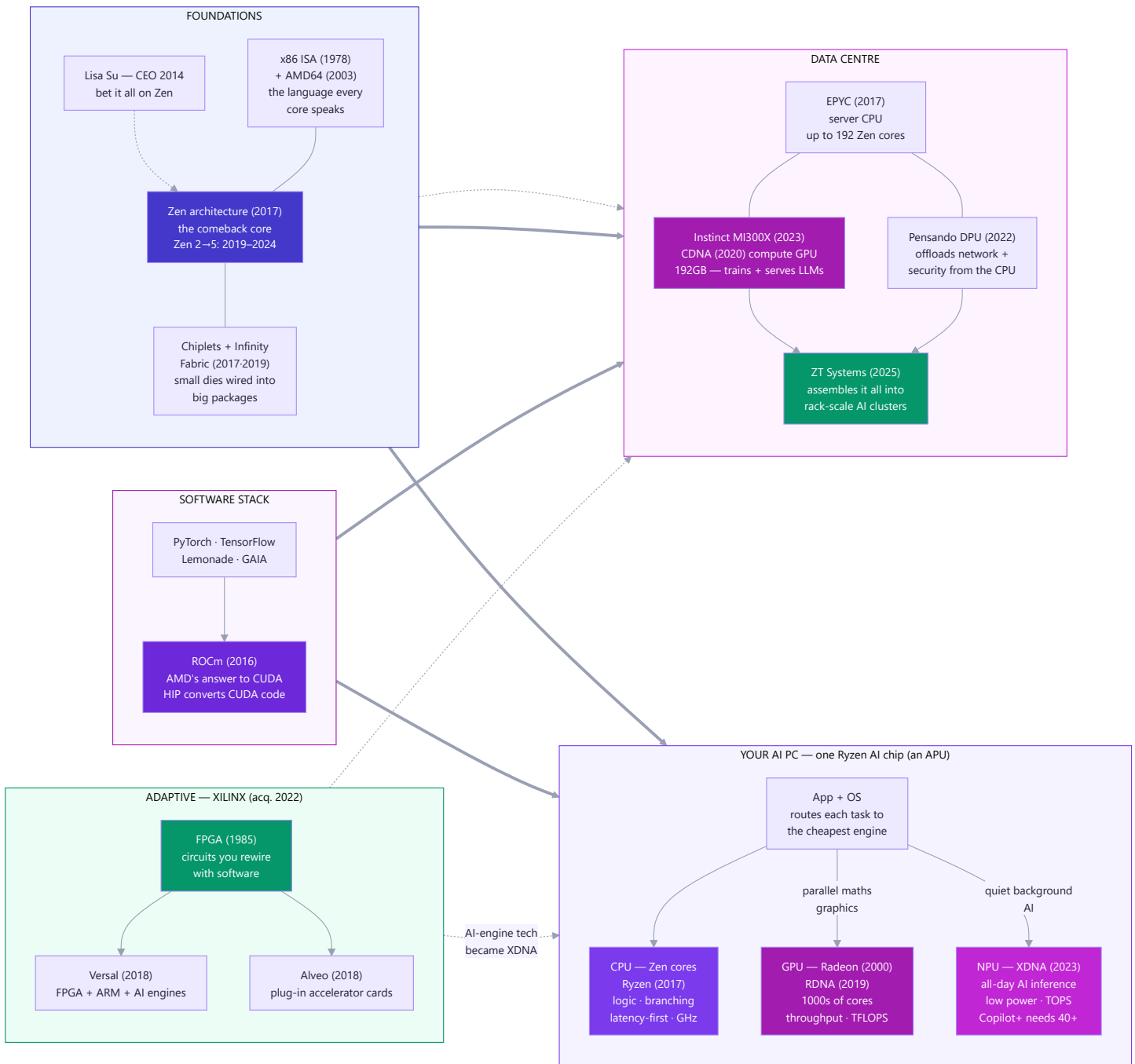
Lisa Su (CEO since 2014)

Dr. Lisa Su took over AMD in **October 2014** when it was near bankruptcy (~\$2/share) and executed one of the great corporate turnarounds in tech history: bet everything on the Zen architecture, re-entered servers with EPYC, bought Xilinx, and pushed AMD into AI — the stock rose over 50× during the run. An MIT-trained electrical engineer, she's the rare CEO who can walk a fab floor and an earnings call with equal credibility. **Example:** her keynotes (CES, Computex, Advancing AI) are where AMD's roadmap is announced — watch one before DevDay and you'll recognise every product in this glossary.

Telugu: Lisa Su 2014 lo CEO ayyindi — AMD daadapu bankrupt, share ~\$2. Zen meeda antha pettubadi, EPYC tho servers loki, Xilinx konnadi, AI loki — tech history lo goppa turnarounds lo okati (stock 50 rettu paiga). MIT electrical engineer; keynotes lo roadmap antha aame prakatistundi. DevDay ki mundu okka keynote chudu.

The Big Picture — every term in one frame

Follow the arrows: the foundations (Zen, AMD64, chiplets) feed every product; your AI PC routes each task to the cheapest engine; one software stack (ROCm) drives both your laptop GPU and the data-centre Instinct; and Xilinx technology flows into the NPU and the racks. Every term from this part appears exactly once.



Telugu: Ee okka bommallo Part M antha undi. **Edama:** foundations — x86/AMD64 bhasha, Zen core design (2017), chiplets + Infinity Fabric; ivi anni products ki punaadi. **Nee AI PC:** okate Ryzen AI chip lo mudu engines — OS prathi pani ni cheapest engine ki pampistundi: logic CPU ki, parallel maths GPU ki, background AI NPU ki. **Madhya:** software stack — PyTorch laantivi ROCm dwara AMD GPUs meeda nadustayi (CUDA ki AMD jawab). **Data centre:** EPYC + Instinct MI300X + Pensando DPU — anni kalipi ZT Systems full racks ga kadutundi. **Kudi:** Xilinx FPGA prapancham — mariyu daani AI-engine technology ye nee laptop lo XDNA NPU ga maarindi. Okka frame lo mottham AMD katha.

How a single AI task flows through this picture: you ask your laptop to summarise a document → the **OS** hands the app logic to the **CPU** (Zen) → the model's matrix maths goes to the **GPU** or, if it's a small quiet model, the **NPU** (XDNA) → the same PyTorch-on-**ROCm** code, scaled up, runs on **Instinct** GPUs beside **EPYC** CPUs in a

ZT-built rack when the model is too big for your lap. One architecture family, one software stack, laptop to data centre — that is AMD's whole pitch, and now the diagram in your head matches the one on their keynote slides.

Prep doc for AMD AI DevDay India 2026 · Taj West End, Bengaluru · Aug 8, 2026. Built from the published agenda + AMD's 2026 stack (Ryzen AI Max+ 395 "Strix Halo", Instinct MI350X, ROCm 7, Lemonade Server, GAIA). Verify session details against the official site closer to the date — agendas shift.